

Library Mobile App for member Self Service

Sri Charan Kuchuru	2520030093
Rishi A	2520030268
Puvvada Sudha Gayathri	2520030455

Under the Guidance
of

Dr. S Spandana

Assitant Professor

Department of Computer Science and Engineering



KONERU LAKSHMAIAH EDUCATION FOUNDATION

GROUP NUMBER:8

Name of Candidate	Role
Sri Charan	Lead
Rishi A	Member
P. Sudha	Member

ABSTRACT

- The Library Mobile App for Member Self-Service is a web-based library management solution designed to make library services faster, easier, and more accessible to students and other library members. The proposed application provides a self-service digital platform through which members can access essential library services using their mobile devices.
- Users can securely log in, search and browse the library catalog, view book details and availability, check their issued books and due dates, request or reserve books, and renew eligible books. The system can also provide notifications for upcoming due dates, overdue books, and other library updates. A backend database stores and manages member, book, borrowing, reservation, and transaction information while maintaining data consistency and security.
- The project applies Relational Database Engineering concepts by designing an Entity-Relationship model and normalized relational schema using MySQL. SQL features such as DDL, DML, joins, subqueries, aggregates, constraints, indexes, views, transactions, and stored logic can be used to maintain data integrity and support efficient operations.
- The system follows RESTful API principles and a layered backend architecture separating routers, services, and repositories. Docker and Docker Compose can be used for containerization, while GitHub supports version control and collaboration.

INTRODUCTION

- The Library Mobile App for Member Self-Service is a mobile application designed to provide students and library members with convenient access to essential library services. Traditional library systems often require members to physically visit the library to search for books, check availability, view borrowed books, or obtain information about due dates.
- The proposed system provides a centralized mobile platform where members can independently search the library catalog, check book availability, view their borrowing history, reserve books, renew eligible books, and receive notifications. The application communicates with a backend server through REST APIs, while a relational database stores and manages member, book, borrowing, and reservation information.
- The project combines mobile application development, database management, and distributed backend development to create an efficient, secure, and scalable library self-service system.

EXISTING SYSTEM

- The existing library system primarily relies on **traditional manual or basic computerized processes** for managing books, members, borrowing, returns, and reservations.
- Members often need to physically visit the library or contact library staff to search for books, check their availability, borrow or renew books, and obtain information about due dates.
- Library staff are responsible for maintaining records and handling most transactions, which can be time-consuming and may lead to data-entry errors or delays in updating information.
- In some systems, book and member records are maintained using spreadsheets or standalone desktop applications, which may not provide real-time access to members. Notifications for due dates and reservations may also be limited or completely manual.
- These limitations make the existing system less convenient, increase the workload of library staff, and restrict members from accessing library services anytime and anywhere.

PROBLEM

Traditional library management systems have several limitations:

- Members often need to visit the library for basic services.
- Searching for books can be time-consuming.
- Members may not know whether a particular book is currently available.
- Manual management of borrowing and reservation records can lead to errors.
- Members may forget book due dates and incur unnecessary fines.
- Library staff spend considerable time handling routine queries.
- Information may not be available to members in real time.
- Separate systems or manual records can make data management difficult.
- Therefore, there is a need for a mobile-based self-service system that provides library information and services to members anytime and anywhere.

SOLUTION(Contribution)

- The proposed Library Mobile App for Member Self-Service provides a digital solution that allows library members to access essential library services through their mobile devices without depending on library staff for routine operations.
- The application enables users to securely log in, search and view books, check real-time availability, reserve books, renew borrowed books, track due dates, and view borrowing history. A distributed backend using Node.js and Express.js handles requests through REST APIs and connects the mobile application with a MySQL database that securely stores member, book, borrowing, reservation, and transaction information.
- The system reduces manual work, improves data accuracy, provides faster access to library services, and makes library operations more efficient. The main contribution of the project is the integration of mobile self-service, database management, and distributed backend technologies into a single scalable and user-friendly library management solution.

Objectives

The main objectives of the project are:

- To develop a mobile-based self-service library application.
- To allow members to search and browse library books easily.
- To provide real-time information about book availability.
- To enable members to reserve and renew books remotely.
- To maintain member, book, borrowing, and reservation data using a relational database.
- To develop RESTful APIs for communication between the mobile application and backend.
- To provide secure authentication and authorization.
- To send reminders about upcoming due dates and reservations.
- To reduce manual work for library staff.
- To design a scalable and reliable backend architecture.

Modules

Module 1: User Registration & Login

- Member registration
- Login/logout
- Password management
- Authentication using JWT
- Role-based access for members and administrators

Module 2: Book Catalog

- View available books
- Search books by title
- Search by author/category
- View book details
- Check availability

Module 3: Book Reservation

- Reserve an available/unavailable book
- View reservation status
- Cancel reservation
- Reservation queue management

Modules

Module 4: Borrowing & Returns

- View currently borrowed books
- View issue date
- View due date
- View borrowing history
- Update return status

Module 5: Book Renewal

- Request renewal through the application
- Check renewal eligibility
- Update due date
- Prevent renewal when rules are violated

Module 6: Notifications

- Due-date reminders
- Overdue notifications
- Reservation availability notifications
- Library announcements

Modules

Module 7: Member Profile

- View personal information
- Update profile details
- View borrowing statistics
- View fines, if implemented

Module 8: Admin/Library Management

- Add/update/delete books
- Manage members
- Manage borrowing transactions
- Manage reservations
- Monitor overdue books
- Generate reports

Module 9: Backend & Database

- REST API implementation
- Database connectivity
- CRUD operations
- Transaction management
- Data validation
- Authentication and authorization

SOFTWARE REQUIREMENTS

- The software requirements for the Library Mobile App for Member Self-Service include Android Studio or Flutter for developing the mobile application, with Dart, Java, or Kotlin as the programming language.
- Node.js with Express.js can be used to develop the distributed backend and RESTful APIs that connect the mobile application with the server.
- MySQL is used as the relational database for storing member, book, borrowing, reservation, and transaction details, while MySQL Workbench can be used for database design and SQL query management. Visual Studio Code can be used for backend development, and Postman can be used to test REST APIs.
- Git and GitHub are used for version control and project collaboration. JWT can be used for secure user authentication and authorization. The application can be tested on Android OS, and platforms such as Render, Railway, or AWS can be used for deploying the backend and database.

CONCLUSION AND FUTURE SCOPE

- The Library Mobile App for Member Self-Service provides a convenient and efficient solution for accessing library services through a mobile device. The system allows members to search for books, check availability, view borrowing details, reserve and renew books, and receive due-date notifications without depending on library staff for routine tasks.
- The integration of a REST-based backend with a MySQL database ensures organized data management, secure communication, and reliable processing of library transactions.
- In the future, the system can be enhanced with QR/barcode or RFID-based book issue and return, AI-based book recommendations, online fine payment, digital e-books, push notifications, voice-based search, analytics dashboards, cloud deployment, and a library chatbot.
- These enhancements can transform the application into a more advanced, automated, and intelligent smart library management system.

THANK YOU